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Reply: Palaeomagnetism, radiometric age and geochemistry of an adamellite at Yetholme, N.S.W.

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Carboniferous poles for Europe form two groups. He implied an apparent polar shift during the Carboniferous since pole pairs, obtained from the same region, often differed by amounts in excess of 20°. Since then, Storetvedt (1967) and McElhinny (1973) have stated similar conclusions. Storetvedt (1967) concentrated on those data that were supported by laboratory stability evidence and was able to show that the two groups also had a clear stratigraphic basis. The difference between the Australian Carboniferous and Permian-Carboniferous groups is about 40°.

These latest results reported by Facer (1976) for southeastern Australia are also consistent with the combined Palaeozoic results for the supercontinent Gondwanaland. The interpretation of the Early Carboniferous pole position for Australia was considered uncertain by McElhinny & Embleton (1974) though they suspected that it belonged to the mid-Palaeozoic group of poles in southern Africa. The pole calculated here from Dr Facer's data confirms that interpretation (see McElhinny & Embleton, 1974, fig. 3—the new pole plots close to the Late Silurian-Early Carboniferous group of poles for Gondwanaland).

The discussion I have presented here has been based on my interpretation of Dr Facer's results. The model used universally by palaeomagnetists to express the long-term average behaviour of the Earth's magnetic field is termed the *axial geocentric dipole field.* A detailed description of the model has been given by Irving (1964) and McElhinny (1973). According to this model, the palaeomagnetic latitude of a sampling site is simply and uniquely derived from the mean inclination according to the relationship:

$$\tan I = 2 \tan \lambda$$

where I = inclination of magnetization and λ = geomagnetic/geographic latitude. Since inclination values vary from one locality to another, it is important to express results in the form of some parameter that has the same value at all observing localities—the parameter used is the palaeomagnetic pole. The pole position computed by Facer (1976) is incorrect and accordingly Figure 3 in that article requires revision.

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REPLY

PALAEOMAGNETISM, RADIOMETRIC AGE AND GEOCHEMISTRY OF AN ADAMELLITE AT YETHOLME, N.S.W.

By R. A. FACER

I am grateful to Dr Embleton for pointing out the error in the calculation of the palaeomagnetic pole position for the Yetholme adamellite given in Facer (1976*b*). Any inconvenience that this error may have caused is regretted.

The Table presents data from Facer's (1976*b*) table IIb, with recalculated pole positions. These results differ by the exclusion or inclusion of the aplite data. Table IIb of Facer (1976*b*) showed that, although the mean direction for the small aplite population ($N = 3$) indicated significance at the 95% level, the mean was not different from the present geomagnetic field at Yetholme. Stability to partial AF demagnetization was not convincingly shown by the aplite. Hence the aplite data were omitted from the original pole calculation. If Dr Embleton's suggestion of a successful baked contact test included the aplite, then some justification is possible for the second pole position (ALL) shown in the Table. This second pole position is closer to other Carboniferous pole positions for Australia (Dr Embleton's figure: Facer, 1976*b*, fig. 3). The Table also contains data using sub-sites (*cf.* Robertson & Watkinson, 1974) as units. The mean direction is very similar to the specimen-unit mean. However, the high palaeolatitude affects the calculated palaeolongitude of the pole.

If Irving's (1966, p. 6025) *'mid-Carboniferous continental drift episode'* is a real 'episode of relative land movement', then the Yetholme data fit into a possible group of Late Devonian-Early Carboniferous poles discrete from a group of Late Carboniferous-Permian poles, and thus lend support to Irving's (1966) suggestion.

Table *Yetholme adamellite palaeogeomagnetic data*

Group (after partial AF demag.)	N	D	I	α_{95}	k	R	Palaeomagnetic Pole			
							Lat.	Long.	dp	dm
ALL (no aplite)	49°	356°T	-47°	6.0°	12.5	45.2	84°S	295°E	5.0°	7.8°
ALL	52°	357°T	-49°	5.9°	12.2	47.8	86°S	289°E	5.6°	7.6°
ALL SITES (no aplite)	5°	359°T	-46°	9.7°	62.7	4.94	84°S	321°E	7.9°	12.4°
ALL SITES	5°	359°T	-46°	10.3°	56.6	4.93	84°S	321°E	8.4°	13.2°

Notes: N is the number of specimens or sites used in the Fisher (1953) analysis to determine the mean direction (declination D, inclination I).

α_{95} is the half-angle of the cone of 95% of confidence, k the estimate of precision and R the length of the resultant vector (Fisherian statistics).

The polar error (dp, dm) defines the oval of 95% confidence in degrees about the pole position.

For ALL SITES (no aplite) A_{95} for the virtual geomagnetic poles is 10.5 degrees, and for ALL SITES A_{95} is 10.8 degrees.

The means discussed here were calculated using PALMAGFISHERANAL (Facer, 1976*a*), and all are significant at the 95% level using the criteria of Watson (1956) and Vincenz & Bruckshaw (1960).

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